IBRAHIM TIJANI, PhD, P.Eng, PMP

Google Scholar | Scopus | Researchgate | Materials informatics | AI for resilient infrastructure

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PROFESSIONAL PROFILE

Civil and structural engineering scholar, registered professional engineer with international academic and industry experience across Canada, Hong Kong, Saudi Arabia, and Nigeria. My research integrates advanced construction materials, structural mechanics, scientific machine learning, explainable artificial intelligence, and data-driven materials modelling for resilient infrastructure. Strong record of peer-reviewed scholarship, externally funded postdoctoral research, interdisciplinary collaboration, and undergraduate teaching across civil engineering and construction disciplines.

RESEARCH INTERESTS

- Advanced construction materials, structural engineering, and rehabilitation.
- Scientific machine learning, generative artificial intelligence, materials informatics, and data-driven modelling.
- Structural health monitoring, non-destructive evaluation, and smart infrastructure systems.
- Sustainable, resilient, and climate-adaptive construction materials and infrastructure.

ACADEMIC APPOINTMENTS

2026 - present

Research Coordinator

School of Engineering, University of British Columbia (UBC), Canada

- Coordinated multidisciplinary research projects, including study planning, experimental activities, data collection, documentation, progress tracking, and preparation of technical reports and research outputs.
- Supported research teams and collaborators by organizing project activities, monitoring milestones, maintaining research records, and facilitating effective communication among stakeholders.
- Contributed to research proposal development, grant applications, manuscript preparation, data analysis, and dissemination of findings through journal publications, technical reports, and conference presentations.

2022 - 2026

Mitacs Accelerate Industrial Postdoctoral Fellow

School of Engineering, University of British Columbia (UBC), Canada

- Conduct research on intelligent and non-destructive solutions for structural and urban infrastructure systems.
- Develop machine-learning frameworks, user-centric engineering tools, and technical reports for research and industry applications.
- Support research partnership development and funding-related activities.

2020 - 2022

Postdoctoral Fellow / Project Coordinator

Department of Building & Real Estate, The Hong Kong Polytechnic University, Hong Kong

- Led and supported projects on smart infrastructure monitoring, leak diagnosis systems, and data-driven engineering applications.
- Coordinated project reporting, research outputs, and grant-related documentation

2017 - 2020

Research and Teaching Assistant / PhD Candidate

Department of Architectural & Civil Engineering, City University of Hong Kong

- Conducted extensive experimental and numerical studies on FRP-repaired and FRP-confined concrete systems.
- Supported undergraduate teaching, laboratory activities, and research dissemination.

2014 - 2017

Teaching Assistant / MSc Student

Civil Engineering Department, King Saud University, Saudi Arabia

- Assisted in teaching, assessment, tutorial support for construction engineering and management courses.

PROFESSIONAL EXPERIENCE

2023 — **Project Engineer / Assessment Officer**
Property Assessment Services, Government of Manitoba, Canada
- Coordinate and execute engineering and infrastructure assessment activities, including site reviews, inspection, valuation-related technical analysis, and reporting.
- Review drawings, technical records, and field information while supporting engineering judgment aligned with relevant codes and standards.
- Mentor and support new staff, interns, and technical workflows.

2022 - 2023 — **Project Engineer in Training**
Spannovation Consulting Ltd., Vancouver Metropolitan Area, BC, Canada
- Reviewed detailed designs, work orders, and project documents for compliance with applicable codes, standards, and regulations.
- Prepared design-check spreadsheets for structural steel components under flexure, shear, compression, and tension actions.

EDUCATION

2024 **Certificate of Completion,** Water Resources Management and Policy, Universite de Geneve.

2020 **Doctor of Philosophy, Civil Engineering**
City University of Hong Kong, Kowloon, Hong Kong SAR
Area of Specialization: Structural Engineering

2017 **Master of Science, Civil Engineering** [GPA: 4.50/5.00]
King Saud University, Riyadh, Saudi Arabia.

2011 **Bachelor of Technology, Civil Engineering** [GPA: 4.17/5.00]
Ladoke Akintola University of Technology, Ogbomoso, Nigeria.

PROFESSIONAL LICENSURE AND CERTIFICATIONS

2023 Registered Professional Engineer, Engineers Geoscientists Manitoba, Canada

2023 Project Management Professional (PMP) Certification

AWARDS, HONORS, AND FELLOWSHIPS

2022 - present — Mitacs Accelerate Industrial Postdoctoral Research Fellowship
School of Engineering, the University of British Columbia (UBC), Canada.

2020 - 2022 — Innovation and Technology Fund Postdoctoral Fellowship
Department of Building & Real Estate, The Hong Kong Polytechnic University

2017 - 2020 — Hong Kong Ph.D. Fellowship Scheme (HKPFS)
Research Grant Council of Hong Kong S.A.R
Department of Architecture & Civil Engineering, City University of Hong Kong

2017 - 2018 — Chow Yei Ching School of Graduate Studies Entrance Scholarships
Department of Architecture & Civil Engineering, City University of Hong Kong

2014 - 2017 — Postgraduate Scholarship, Ministry of Education
Civil Engineering Department, King Saud University

TEACHING EXPERIENCE

2020 - 2022

Postdoctoral Fellow, The Hong Kong Polytechnic University, Hong Kong SAR

- Courses taught / supported: Project Management; Construction Administration; Engineering Surveying & Photogrammetry; Quantity Surveying; Construction Law; Applied Research Project; Survey Computations.
- Responsibilities included course preparation, student evaluation, lesson planning, grading, supervision of final-year research projects, and curriculum support

2017 - 2020

Research and Teaching Assistant, City University of Hong Kong, Hong Kong SAR

- Courses taught / supported: Structural Analysis; Fluid Mechanics; Engineering Geology; Steel Design; Hydraulics and Hydrology; Foundation Engineering; Geotechnical Engineering; Engineering Drawing; Project Administration; Highway Design; Computer Applications in Civil Engineering; Construction Management.
- Supported course delivery, tutorials, grading, and coordination of teaching activities.

2014 - 2017

Teaching Assistant, King Saud University, Saudi Arabia

Courses supported: Construction Engineering and Management; Construction Planning and Control; Cost Analysis and Control.

PEER-REVIEWED JOURNAL PUBLICATIONS

24 peer-reviewed journal articles | 17 Q1 | 6 Q2 | 23/24 (96%) published in Q1 – Q2 journals

1. **Tijani I.A**, Haroglu H., Wakjira T.G, Abadel A., & Alam M.S (2026). Data-augmented explainable machine learning framework and user-centric tool for classifying failure modes and predicting flexural capacity of FRP-strengthened RC beams. *Structures*. 111776. [Q1].
2. **Tijani I.A**, Kanaan D, Wakjira T.G, Alam M.S, & Haroglu H (2026). Explainable machine learning in concrete technology: A state-of-the-art review with future innovations. *Archives of Computational Methods in Engineering*. 33, 6095-6123. [Q1].
3. **Tijani I.A** & C Jiang. (2025) GUI-based hybrid ML model for predicting ultimate strength of FRP-confined UHPC with CTGAN-augmented data. *Composite Structures*. 119905. [Q1].
4. **Tijani I.A**, T.G Wakjira, M.S Alam, & N Uddin. (2025) Digital image correlation for structural health monitoring of bridge systems: A state-of-the-art review with future research directions. *Archives of Computational Methods in Engineering*. 33, 4325-4341. [Q1].
5. Zhang R, Fares A., **Tijani I.A**, Zayed T. Jin Z, and Yussif A. (2025) Acoustic based leak location detection for water supply pipelines in urban areas via multi-task deep learning. *Engineering Applications of Artificial Intelligence*. 1112718. [Q1].
6. **Tijani I.A**, T.G Wakjira, H. Haroglu & M.S Alam. (2025). A surrogate-based machine learning model for prediction of interfacial fracture energy in externally bonded FRP–concrete bonds with a user-friendly tool. *Engineering Fracture Mechanics*. 111633. [Q1].
7. **Tijani I.A**, T.G Wakjira, H. Haroglu & M.S Alam. (2025) Explainable machine learning and application-oriented tool for predicting effective hoop strain of FRP-confined concrete. *Frontiers of Structural and Civil Engineering*. 19, 1621-1636. [Q1].
8. Wakjira, T.G., **Tijani, I.A**., Alam, M.S., Mashal, M., & Hasan, M.K. (2025) Can we trust AI content detection tools for critical decision-making? *Information*, 16, 904. [Q2].
9. Liu, R., Tariq, S., **Tijani, I.A**, Fan, H, Abdelmageed, S Fares, A., & Zayed, T. (2025). Leak detection in water distribution networks using micro-electromechanical systems-based accelerometers: a machine learning approach. *Water Practice and Technology*. 20(9), 1900-1920. [Q2].
10. **Tijani I.A**, J.A Dauda, M.A Kareem, & A.I Lawal. (2025). Genetic programming-based model for estimating maximum pull load of FRP-to-concrete bond interfaces with GUI implementation. *Structural Concrete*. 26(6), 8899-8912 [Q1].
11. Zhang, R., Yussif, A., **Tijani, I.A**., Fares, A., Tariq, S., & Zayed, T (2024). Acoustic localization approach for urban water distribution networks using machine learning method. *Engineering Applications of Artificial Intelligence*. 137. 109062. [Q1].
12. Liu, R., Tariq, S., **Tijani, I.A**., Fares, A., Bakhtawar, B., Fan, H., Zhang, R., & Zayed, T. (2024). Data-driven approaches for vibroacoustic localization of leaks in water distribution networks. *Environmental Processes.* 11, 14. [Q1].
13. **Tijani, I.A**., Lawal, A.I., Ogunsola, N.O, & Kwon, S. (2023) Prediction of ultimate strength of FRP-confined predamaged concrete using backward multiple regression motivated soft computing methods. *Scientia Iranica*. e23225. [Q3].

14. Kareem, M.A., Akintonde, B.B., Adesoye, J.S., Ishola, K., **Tijani, I.A.**, & others. (2023). Influence of Cashew Leaf Ash as Partial Replacement for Cement on the Properties of Fresh and Hardened Concrete. *Cleaner Waste Systems*. 4, 100063. [Q2].
15. **Tijani, I.A.**, Lawal, A.I., & Kwon, S. (2022). Machine learning techniques for prediction of ultimate strain of FRP-confined concrete. *Structural Engineering and Mechanics* 84(1):101-111 [Q2].
16. **Tijani I.A** & Zayed T. (2022). Gene Expression Programming Based Mathematical Models for Leak Detection of Water Distribution Networks. *Measurement*, 188:110611 [Q1].
17. Dauda, J.A., Rahmon, S., **Tijani, I.A.,** & Mohammed, F. (2022). Design Optimisation of Reinforced Concrete Pile Foundation using Generalized Reduced Gradient Algorithm. *Frontiers in Engineering and Built Environment* 2(3):133-153. [Q2].
18. **Tijani I.A**., Abdelmageed, S., Fares, A., Fan, K.H., Hu, Z.Y., & Zayed, T., (2022). Improving the leak detection efficiency in water distribution networks using noise loggers. *Science of the Total Environment*. 821, 153530. [Q1].
19. Fares, A., **Tijani, I. A** & others (2022) Leak detection in real water distribution networks based on acoustic emission and machine learning. *Environmental Technology*. 44(25), 3850-3866. [Q2].
20. **Tijani I.A**., Jiang, C., Lim. C.W., & Wu, Y.F. (2021). Eccentrically loaded concrete under nonuniform passive confinement. *ASCE Journal Structural Engineering*. 04021247. [Q1].
21. **Tijani I.A**., Wu, Y.F., & Lim, C.W. (2021). "Effects of pre-damage on stress-strain relationship of partially confined concrete." *ACI Structural Journal*. 118(1):61–72. [Q1].
22. **Tijani I.A**., Jiang, C., Lim. C.W., & Wu, Y.F. (2020). Effect of load eccentricity on the mechanical response of FRP-confined pre-damaged concrete columns. *ASCE Journal Composites for Construction*, 04020057. [Q1].
23. **Tijani I.A** & others (2020). "Energy balance method for modeling ultimate strain of fiber-reinforced polymer-repaired concrete." *Structural Concrete*, 21(2):804–820. [Q1].
24. **Tijani I. A** & others (2019). "Aggregate size effects and general static loading response on mechanical behavior of passively confined concrete." *Construction and Building Materials*, 205:61–72. [Q1].

CONFERENCE PROCEEDINGS AND PROFESSIONAL SERVICE

- Refereed conference contributions include APFIS 2019, EMI 2019, HKSTAM 2019, the First International Conference on Mechanics of Advanced Materials and Structures, and SDSV 2017.
- Technical reviewer for Engineering Structures, Structural Concrete, Construction and Building Materials, Journal of Mechanics of Materials and Structures, Acta Geophysica, Applied Acoustics, Canadian Journal of Civil Engineering, and Nature Communications.

SELECTED RESEARCH PROJECTS

2022 - present	Precast system with high-strength lightweight green concrete, Supervisor: Prof. Shahria Alam *Sponsor: Mitacs Accelerate, Canada*.
2020 - 2022	Smart noise logger-based leak diagnosis system, Supervisor: Prof. Tarek Zayed *Sponsor: Innovation and Technology Fund, Hong Kong SAR*
2017 - 2020	Mechanical relationship of concrete with uniform and non-uniform confinement, Supervisors: Prof. C.W. Lim, Prof. Yu-Fei Wu, Dr. Cheng Jiang *Research grant funded, Hong Kong SAR*

PROFESSIONAL SERVICE

- Technical reviewer for journals including Engineering Structures, Structural Concrete, Construction and Building Materials, Journal of Mechanics of Materials and Structures, Acta Geophysica, Applied Acoustics, Canadian Journal of Civil Engineering, and Nature Communications.